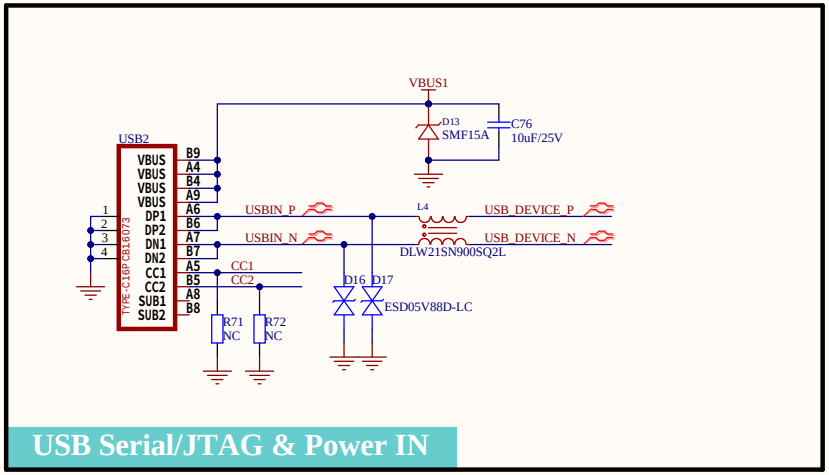
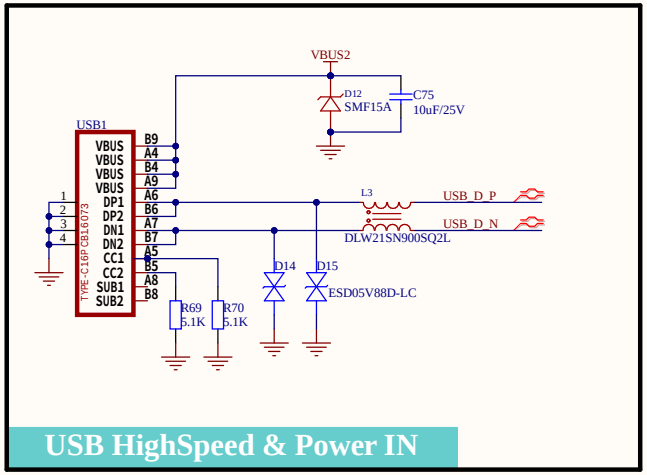
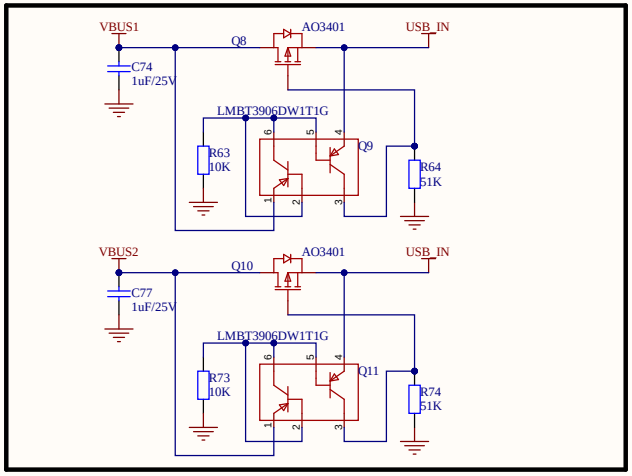
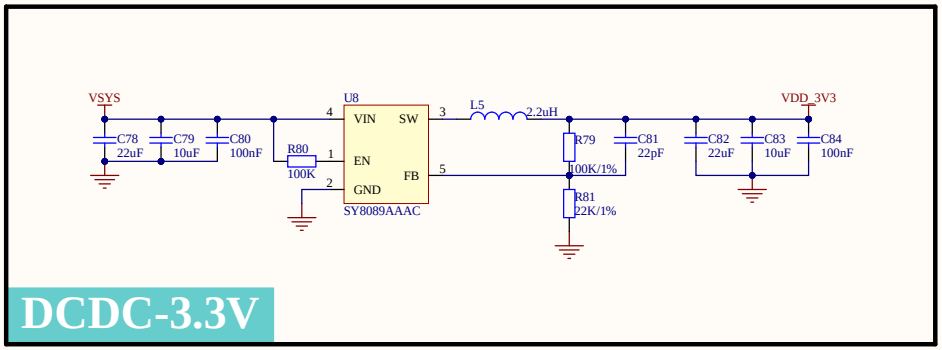
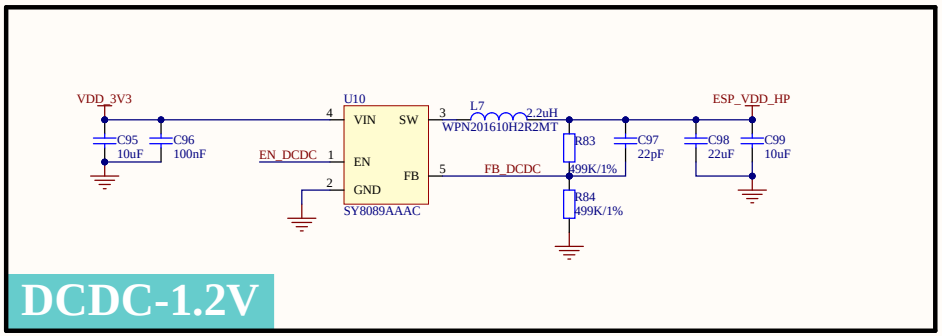
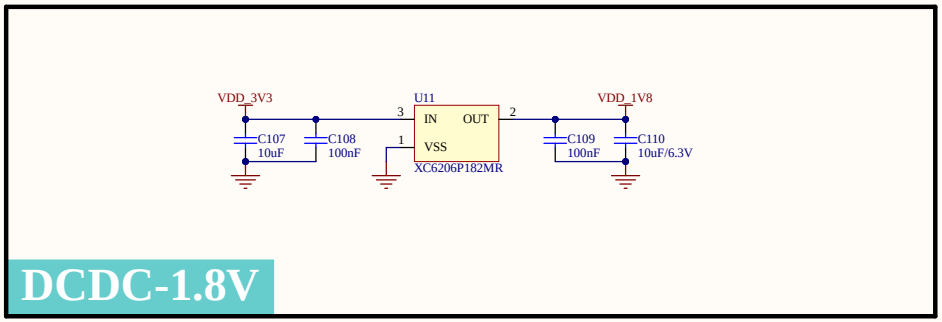

USB2
VBUS1
D13
SMF15A
C76
10uF/25V
USBIN_P
USBIN_N
L4
USB_DEVICE_P
USB_DEVICE_N
DLW21SN900SQ2L
CC1
CC2
D16
D17
ESD05V88D-LC
R71
NC
R72
NC
USB Serial/JTAG & Power IN

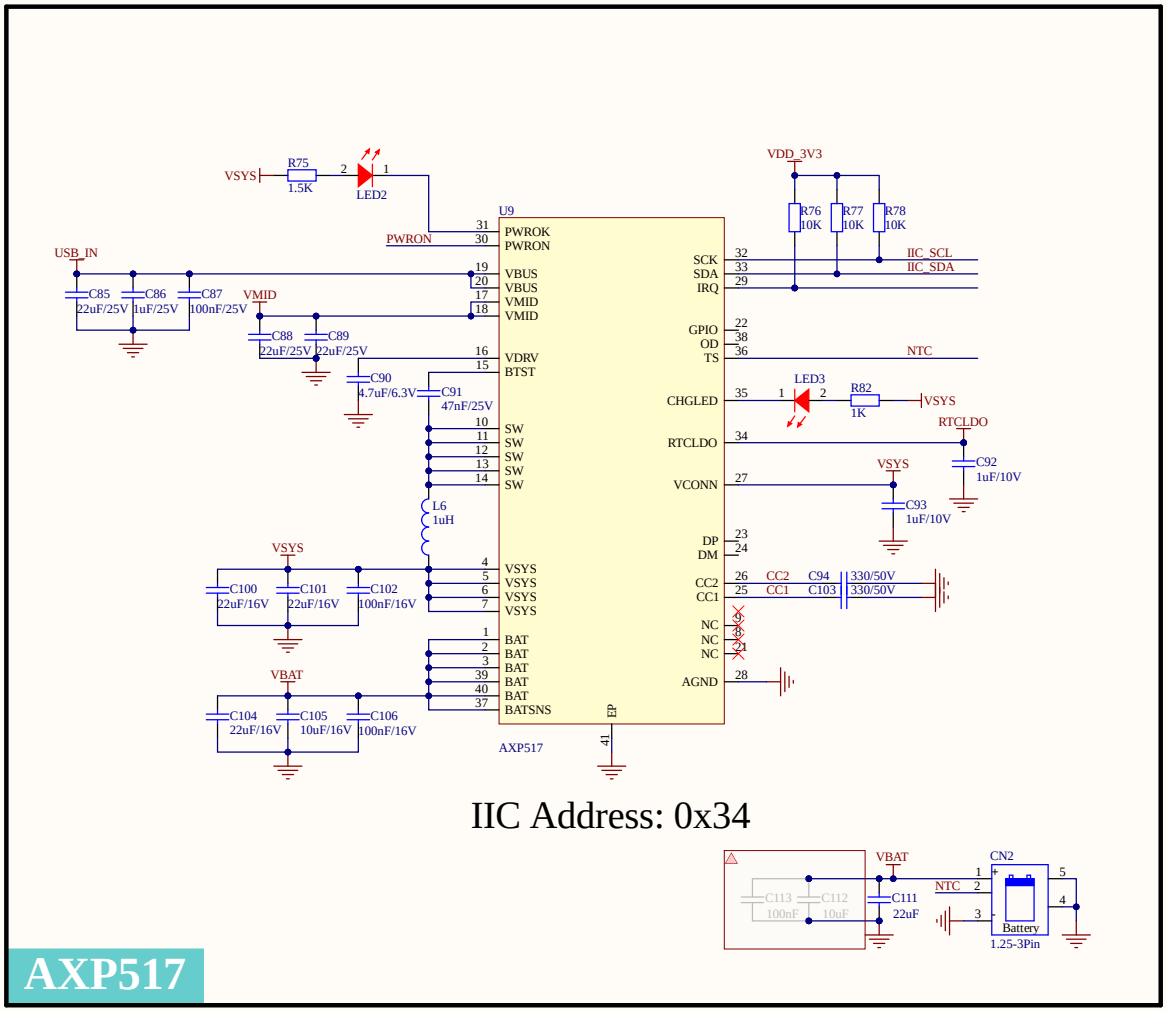

USB1
VBUS2
D12
SMF15A
C75
10uF/25V
L3
USB_D_P
USB_D_N
DLW21SN900SQ2L
D14
D15
ESD05V88D-LC
R69
5.1K
R70
5.1K
USB HighSpeed & Power IN

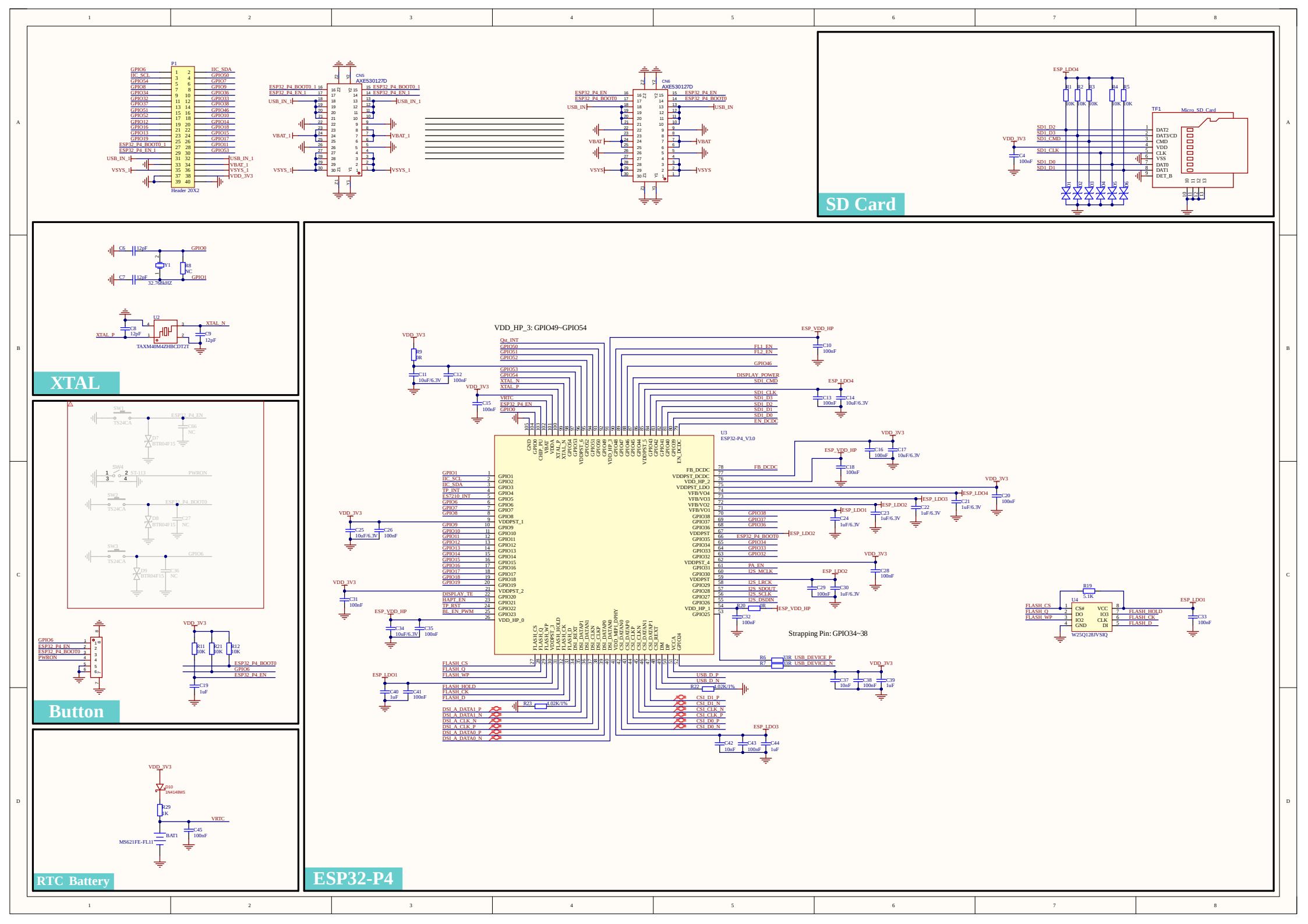

VBUS1
Q8
AO3401
USB_IN
C74
1uF/25V
LMBT3906DW1T1G
R63
10K
Q9
R64
51K
VBUS2
Q10
AO3401
USB_IN
C77
1uF/25V
LMBT3906DW1T1G
R73
10K
Q11
R74
51K

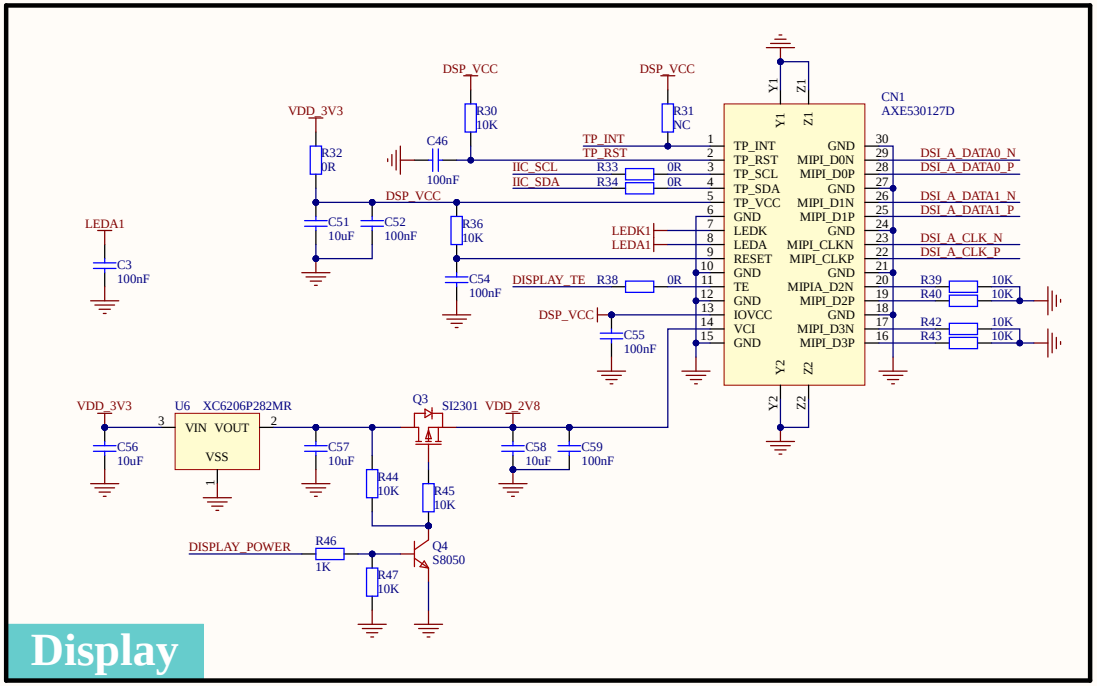

VSYS
U8
VIN
SW
EN
FB
GND
SY8089AAAC
L5
2.2uH
VDD_3V3
C78
22uF
C79
10uF
C80
100nF
R80
100K
R79
100K/1%
C81
22pF
R81
22K/1%
C82
22uF
C83
10uF
C84
100nF
DCDC-3.3V

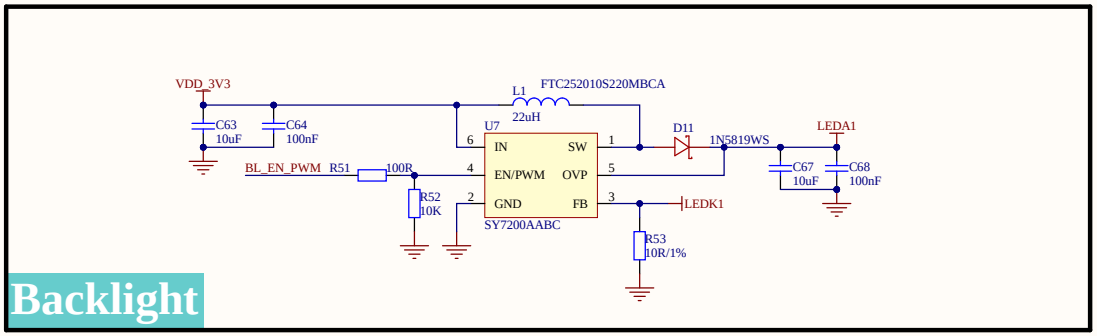

VDD_3V3
U10
VIN
SW
EN
FB
GND
SY8089AAAC
L7
WPN201610H2R2MT
2.2uH
ESP_VDD_HP
C95
10uF
C96
100nF
EN_DCDC
R83
499K/1%
C97
22pF
FB_DCDC
R84
499K/1%
C98
22uF
C99
10uF
DCDC-1.2V

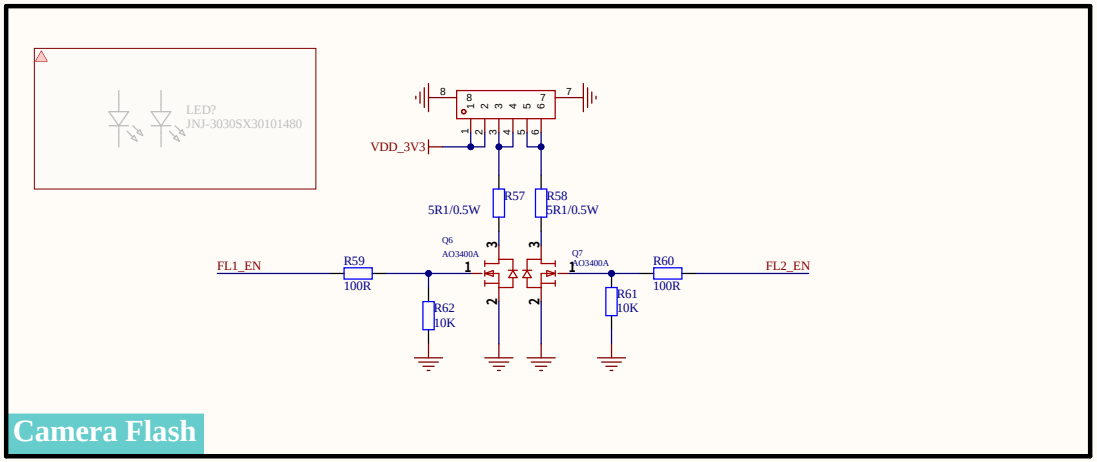

VDD_3V3
U11
IN
OUT
VSS
XC6206P182MR
VDD_1V8
C107
10uF
C108
100nF
C109
100nF
C110
10uF/6.3V
DCDC-1.8V

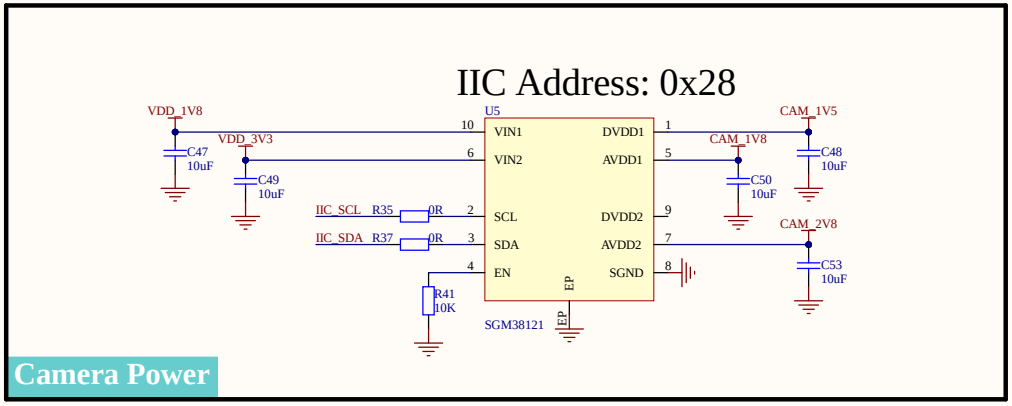

VSYS
R75
1.5K
LED2
VDD_3V3
R76
10K
R77
10K
R78
10K
U9
PWROK
PWRON
IIC_SCL
IIC_SDA
USB_IN
C85
22uF/25V
C86
1uF/25V
C87
100nF/25V
VMID
C88
22uF/25V
C89
22uF/25V
VBUS
VMID
VDRV
BTST
C90
4.7uF/6.3V
C91
47nF/25V
SW
L6
1uH
SCK
SDA
IRQ
GPIO
OD
TS
NTC
CHGLED
LED3
R82
1K
VSYS
RTCLDO
C92
1uF/10V
VCONN
C93
1uF/10V
DP
DM
CC2
CC1
C94
330/50V
C103
330/50V
NC
AGND
C100
22uF/16V
C101
22uF/16V
C102
100nF/16V
BAT
BATSNS
VBAT
C104
22uF/16V
C105
10uF/16V
C106
100nF/16V
EP
AXP517
IIC Address: 0x34
C113
100nF
C112
10uF
C111
22uF
CN2
NTC
Battery
1.25-3Pin
AXP517

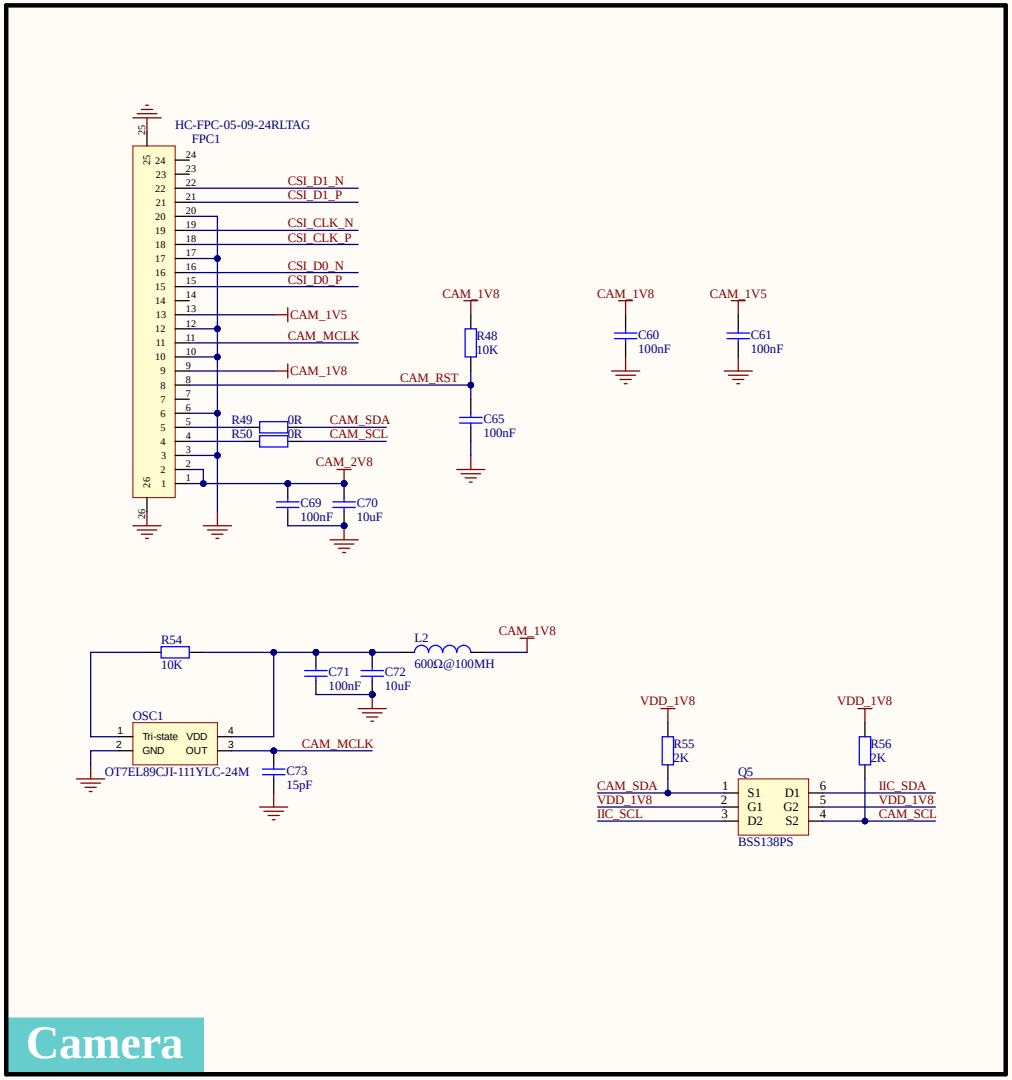

P1
Header 20X2
CN5
AXE530127D
CN6
AXE530127D
SD Card
ESP_LDO4
TF1
Micro_SD_Card
XTAL
U2
TAXM40M4ZHBCDT2T
Button
RTC Battery
BAT1
MS621FE-FL11
ESP32-P4
VDD_HP_3: GPIO49~GPIO54
U3
ESP32-P4_V3.0
Strapping Pin: GPIO34~38
U4
W25Q128JVSIQ

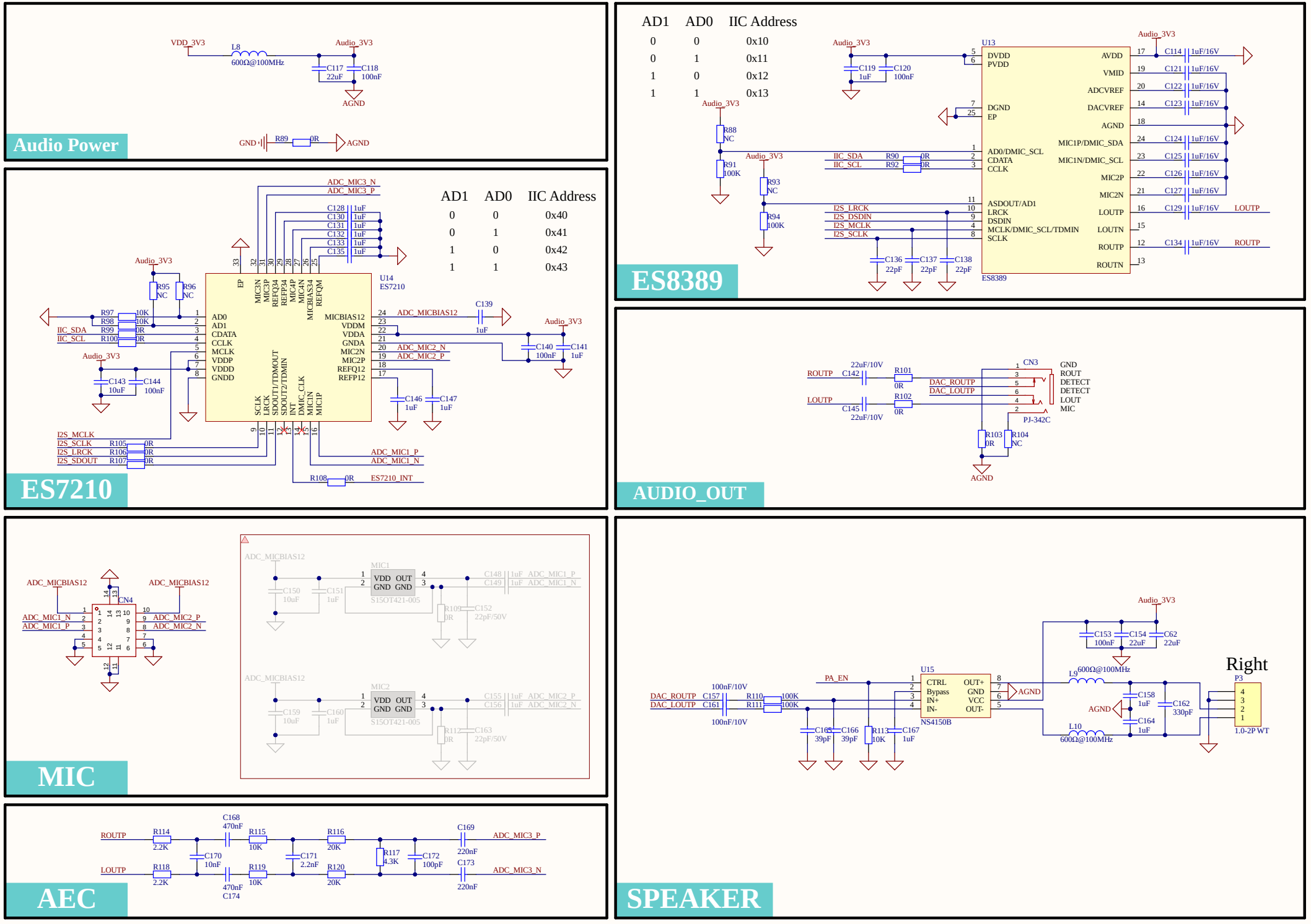

DSP_VCC
DSP_VCC
VDD_3V3
R30
10K
R31
NC
R32
0R
C46
100nF
TP_INT
TP_RST
IIC_SCL
IIC_SDA
R33
R34
0R
0R
DSP_VCC
LEDA1
C3
100nF
C51
10uF
C52
100nF
R36
10K
C54
100nF
LEDK1
LEDA1
DISPLAY_TE
R38
0R
DSP_VCC
C55
100nF
CN1
AXE530127D
TP_INT
TP_RST
TP_SCL
TP_SDA
TP_VCC
GND
LEDK
LEDA
RESET
GND
TE
GND
IOVCC
VCI
GND
GND
MIPI_D0N
MIPI_D0P
GND
MIPI_D1N
MIPI_D1P
GND
MIPI_CLKN
MIPI_CLKP
GND
MIPIA_D2N
MIPI_D2P
GND
MIPI_D3N
MIPI_D3P
DSI_A_DATA0_N
DSI_A_DATA0_P
DSI_A_DATA1_N
DSI_A_DATA1_P
DSI_A_CLK_N
DSI_A_CLK_P
R39
R40
R42
R43
10K
10K
10K
10K
VDD_3V3
U6 XC6206P282MR
VIN VOUT
VSS
C56
10uF
C57
10uF
Q3
SI2301
VDD_2V8
C58
10uF
C59
100nF
R44
10K
R45
10K
DISPLAY_POWER
R46
1K
Q4
S8050
R47
10K
Display

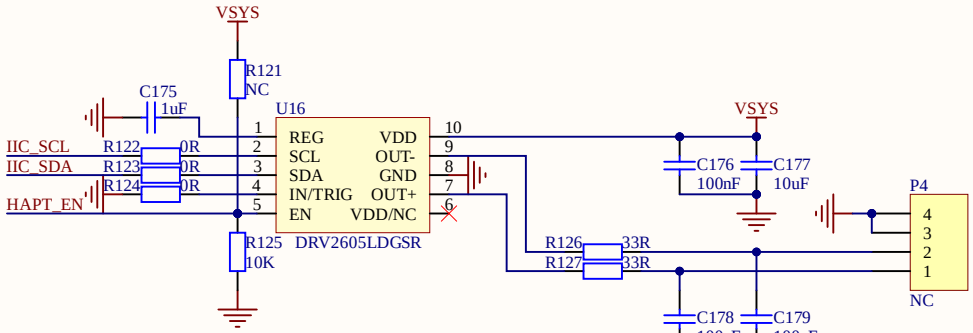

VDD_3V3
L1
FTC252010S220MBCA
22uH
C63
10uF
C64
100nF
U7
IN
SW
EN/PWM
OVP
GND
FB
SY7200AABC
D11
1N5819WS
LEDA1
BL_EN_PWM
R51
100R
R52
10K
C67
10uF
C68
100nF
LEDK1
R53
10R/1%
Backlight


LED?
JNJ-3030SX30101480
VDD_3V3
5R1/0.5W
R57
R58
5R1/0.5W
Q6
AO3400A
Q7
AO3400A
FL1_EN
R59
100R
R60
100R
FL2_EN
R61
10K
R62
10K
Camera Flash


IIC Address: 0x28
VDD_1V8
VDD_3V3
C47
10uF
C49
10uF
U5
VIN1
VIN2
SCL
SDA
EN
DVDD1
AVDD1
DVDD2
AVDD2
SGND
EP
SGM38121
IIC_SCL
IIC_SDA
R35
R37
0R
0R
R41
10K
CAM_1V5
CAM_1V8
CAM_2V8
C48
10uF
C50
10uF
C53
10uF
Camera Power

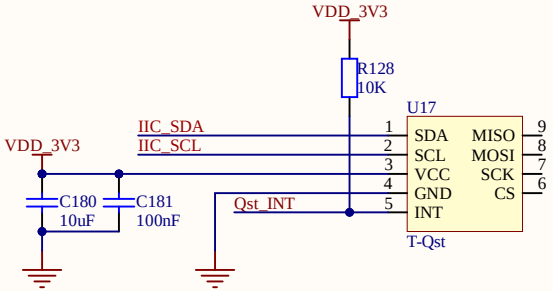

HC-FPC-05-09-24RLTAG
FPC1
CSI_D1_N
CSI_D1_P
CSI_CLK_N
CSI_CLK_P
CSI_D0_N
CSI_D0_P
CAM_1V5
CAM_MCLK
CAM_1V8
CAM_RST
CAM_2V8
R48
10K
C65
100nF
R49
R50
0R
0R
CAM_SDA
CAM_SCL
C69
100nF
C70
10uF
C60
100nF
C61
100nF
R54
10K
L2
600Ω@100MH
C71
100nF
C72
10uF
OSC1
Tri-state
VDD
GND
OUT
OT7EL89CJI-111YLC-24M
C73
15pF
R55
2K
R56
2K
Q5
S1
D1
G1
G2
D2
S2
BSS138PS
CAM_SDA
VDD_1V8
IIC_SCL
IIC_SDA
VDD_1V8
CAM_SCL
Camera


Audio Power
VDD_3V3
L8
600Ω@100MHz
Audio_3V3
C117 22uF
C118 100nF
AGND
GND
R89 0R
AGND
ES7210
AD1 AD0 IIC Address
0 0 0x40
0 1 0x41
1 0 0x42
1 1 0x43
ADC_MIC3_N
ADC_MIC3_P
C128 1uF
C130 1uF
C131 1uF
C132 1uF
C133 1uF
C135 1uF
U14
ES7210
Audio_3V3
R95 NC
R96 NC
R97 10K
R98 10K
R99 0R
R100 0R
IIC_SDA
IIC_SCL
AD0
AD1
CDATA
CCLK
MCLK
VDDP
VDDD
GNDD
MICBIAS12
VDDM
VDDA
GNDA
MIC2N
MIC2P
REFQ12
REFP12
ADC_MICBIAS12
C139 1uF
C140 100nF
C141 1uF
C143 10uF
C144 100nF
C146 1uF
C147 1uF
I2S_MCLK
I2S_SCLK
I2S_LRCK
I2S_SDOUT
R105 0R
R106 0R
R107 0R
ADC_MIC1_P
ADC_MIC1_N
R108 0R
ES7210_INT
ES8389
AD1 AD0 IIC Address
0 0 0x10
0 1 0x11
1 0 0x12
1 1 0x13
Audio_3V3
C119 1uF
C120 100nF
U13
ES8389
DVDD
PVDD
DGND
EP
AD0/DMIC_SCL
CDATA
CCLK
ASDOUT/AD1
LRCK
DSDIN
MCLK/DMIC_SCL/TDMIN
SCLK
AVDD
VMID
ADCVREF
DACVREF
AGND
MIC1P/DMIC_SDA
MIC1N/DMIC_SCL
MIC2P
MIC2N
LOUTP
LOUTN
ROUTP
ROUTN
C114 1uF/16V
C121 1uF/16V
C122 1uF/16V
C123 1uF/16V
C124 1uF/16V
C125 1uF/16V
C126 1uF/16V
C127 1uF/16V
C129 1uF/16V
C134 1uF/16V
R88 NC
R91 100K
R93 NC
R94 100K
R90 0R
R92 0R
I2S_LRCK
I2S_DSDIN
I2S_MCLK
I2S_SCLK
C136 22pF
C137 22pF
C138 22pF
AUDIO_OUT
ROUTP
LOUTP
C142 22uF/10V
C145 22uF/10V
R101 0R
R102 0R
DAC_ROUTP
DAC_LOUTP
CN3
PJ-342C
GND
ROUT
DETECT
DETECT
LOUT
MIC
R103 0R
R104 NC
AGND
MIC
ADC_MICBIAS12
CN4
ADC_MIC1_N
ADC_MIC1_P
ADC_MIC2_P
ADC_MIC2_N
MIC1
MIC2
VDD OUT
GND GND
S15OT421-005
C150 10uF
C151 1uF
C148 1uF
C149 1uF
R109 0R
C152 22pF/50V
C159 10uF
C160 1uF
C155 1uF
C156 1uF
R112 0R
C163 22pF/50V
SPEAKER
Audio_3V3
C153 100nF
C154 22uF
C62 22uF
U15
NS4150B
PA_EN
CTRL
Bypass
IN+
IN-
OUT+
GND
VCC
OUT-
AGND
DAC_ROUTP
DAC_LOUTP
C157 100nF/10V
C161 100nF/10V
R110 100K
R111 100K
C165 39pF
C166 39pF
R113 10K
C167 1uF
L9 600Ω@100MHz
L10 600Ω@100MHz
C158 1uF
C164 1uF
C162 330pF
Right
P3
1.0-2P WT
AEC
ROUTP
LOUTP
R114 2.2K
R118 2.2K
C168 470nF
C174 470nF
C170 10nF
R115 10K
R119 10K
C171 2.2nF
R116 20K
R120 20K
R117 4.3K
C172 100pF
C169 220nF
C173 220nF
ADC_MIC3_P
ADC_MIC3_N



MOTOR

IIC Address: 0x5A



QMC6309 IIC Address: 0x7C

QMI8658C IIC Address: 0x6A or 0x6B

QMI8658C and QMC6309